# Quiz 8 review Math 2270

**Ex 1** Show that similarity of matrices ($B \sim A$ if $B = M^{-1}AM$ for some inv. matrix $M$) is an equivalence relation.

**Ex 2** Show that if $B \sim A$ then $A$ & $B$ have the same eigenvalues and the eigenvectors of $A$ are in 1-1 correspondence with the eigenvectors of $B$.

**Ex 3** Find the JCF of $\begin{pmatrix} 5 & 1 \\ -1 & 3 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

**Ex 4** Let $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ compute $J^2, J^3, J^K, J^{-1}$

**Ex 5** Compute $\begin{pmatrix} 5 & 1 \\ -1 & 3 \end{pmatrix}^K$ and solve $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 5 & 1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ $\quad x(0) = 1$, $y(0) = 2$

**Ex 6** Which of these matrices are similar

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

**Ex 7** Give the definition of linear transformation.

**Ex 8** Write the matrices corresponding to the following linear transformations

$\mathbb{R}^2 \to \mathbb{R}^2$ $\quad T(v_1, v_2) = (v_2, v_1)$; $\quad T(v_1, v_2) = \begin{pmatrix} v_1 + v_2 \\ v_1 - v_2 \\ v_2 \end{pmatrix}$ $\quad \mathbb{R}^2 \to \mathbb{R}^3$

$\mathbb{R}^3 \to \mathbb{R}$ $\quad T(v_1, v_2, v_3) = (v_1, -v_1, 0)$; $\quad T(v_1, v_2, v_3) = (v_1 + v_2, v_2 + v_3, v_3)$

**Ex 9** Write the linear transf corresponding to rotation by $\alpha$ in $\mathbb{R}^2$ (reflection by $y$-axis, reflection by center) and find its matrix.